Building A Furnace

Environmental, Health & Safety Revision | DRAFT Rev L

SAFETY & PPE - MANDATORY

MATERIAL HAZARD NOTICE: This procedure exposes workers to refractory ceramic fiber (RCF), amorphous silica, Portland cement dust, and caustic cleaners. RCF is classified as a suspected carcinogen by inhalation (IARC Group 2B). Cutting, grinding, or disturbing insulation board and cement generates respirable dust.

RESPIRATORY PROTECTION (MANDATORY):

All personnel who cut insulation board, mix/pour cement, or work in the dust zone MUST wear a NIOSH-approved P100 particulate respirator (N100/P100), properly fitted and seal-checked. N95 is NOT sufficient for cutting operations. Reference Section 8 of the RCF SDS (Ceramic Fiber Board LD 2300 F).

ENGINEERING CONTROLS (MANDATORY):

Cut insulation board with the saw connected to a HEPA H-class vacuum (99.97% @ 0.3 micron). Use local exhaust / downdraft station. Do NOT use compressed air to clean dust. Clean up with HEPA vacuum or wet sweep only.

ADDITIONAL PPE:

Safety glasses with side shields (or goggles for caustic work); chemical-resistant gloves; long sleeves; washable or disposable coveralls when handling fiber board or caustic; steel-toe shoes. Follow the caustic SDS for dilution and handling.

HOUSEKEEPING:

Vacuum or wet-sweep the work area frequently. Never dry-sweep or use compressed air. Dispose of waste fiber/cement per applicable regulations. Do not take unwashed work clothing home.

OBJECTIVE

Define the method of building a completed furnace, incorporating required safety controls for hazardous dust and chemical exposure.

RESPONSIBILITY

- The Maintenance Team and Furnace Tech are responsible for new furnace building, and for installing and removing furnaces.
- The Maintenance Team and Furnace Tech are responsible for using the required PPE and engineering controls identified above.
- Supervisors are responsible for ensuring PPE is worn and dust controls are in place prior to and during work.

PREP WORK

1. Soak furnace plates in caustic bath for approximately 48-72 hrs. Upon removal clean off using Borax soap. Wear chemical-resistant gloves and goggles during caustic handling.
2. Inspect furnace liners for cracks, dents-clean outside.
3. Inspect furnace frames for cracks, welds-patch work on bottom.
4. Furnace plate will have already been through inspection.
5. With bottom of the furnace exposed the plate can now be installed. Using the grout bag put a bead of "green patch 421" around the recess in the liner, center the plate in the recess and press it down.
6. Cut and install the 1/4" "Kao Tex Rope" around the outside edge of the plate (the rope is to be covered in "green patch" prior to installation).
7. Continue to smooth and trowel the "green patch" over the bottom of the exposed liner and the exposed edges of the plate. Allow this "green patch" to dry for 24 hours. If cracks are noted, it will need recoated.

PROCESS

1. Prepare the furnace frame for the liner.
2. Inspect the furnace frame for cracks, broken welds, bent framework, and/or missing parts. Inspect for any cracks visible on both sides of the liner, any cracks greater than 4" long or 1/32" wide-along with any crack near the probe tube or end cap will trigger rejection.
3. After approval of the frame fill the bottom cavities with mortar and 1/2" insulation board.
4. After the bottom cavities are filled with mortar, cut and install 1/2" insulation board in the bottom of the frame.
5. After the 1/2" insulation board is installed in the bottom cut the 2" insulation board to go on the sides and the front and back, the front piece is to have a 5" hole cut into it to go over the burner port of the liner. [WEAR P100 RESPIRATOR AND USE HEPA DUST-EXTRACTING SAW WHEN CUTTING BOARD.]
6. Lower the liner into the frame centering it from side to side and end to end.

DRAFT FOR REVIEW - NOT YET RELEASED. Do not use for production until approved.

7. Use 1/2" insulation board strips to fill the gap between the liner and the plate opening in the frame. [WEAR P100 RESPIRATOR WHEN HANDLING/CUTTING BOARD.]
8. To help ensure alignment of liner use small blocks to help keep centered.
9. Recheck the liner for being centered in the box built from the insulation board and recheck liner relation to the frame.
10. Place inserts into the feeder port and thermocouple port in preparation of pouring cement.
11. With the liner in place mix the "Delta T Crete 26 or 30 M" to a consistency of a thin soup. [WEAR P100 RESPIRATOR AND EYE PROTECTION WHEN MIXING/POURING.] Pour the mix along both sides filling the voids along the liner on sides, front and back with the "Delta Crete mix", continue to mix and pour till the burner port is under the cement.
12. Finishing filling the furnace with a thick mix that can be troweled and finish around the feeder hole and formed around the thermocouple hole (plate).
13. Allow furnace to set overnight before proceeding.
14. After the furnace has set attach two metal straps across the top of the furnace to keep it in the cradle as it is being rolled.
15. After the "green patch" has set for 24 hours the bottom can be finished with "Delta T Crete 26 or 30 M". Mix to a stiff consistency, stuff the void between the liner and the frame. Completely filling the void and troweling to the edge of the recess in the plate back to the edge of the opening in the frame.
16. Allow the furnace to set for 24 hours for drying.
17. After the furnace has set roll the furnace back over using the hoist and straps.
18. The furnace is now prepared and will need a minimum of a 14-day dry time. If the furnace has a less than 14-day minimum dry time the supervisor must approve release.
19. Write Pour date, Plate type and Furnace # on the finished furnace.
20. Form 5.9.92 Furnace History Log sheet will need to be completed for each furnace.

PROPOSED REVISION (L)

DRAFT - Added MANDATORY Safety & PPE section to address RCF/silica respiratory exposure from cutting insulation board and mixing/pouring cement, per SDS hazard review. Pending air sampling, engineering control (HEPA dust-extracting saw), and management approval.